

Data Visualization and Proximity

1 Motivation: Why Visualize Data?

Data visualization maps data onto **graphical primitives** to gain insight into information spaces.

- **Qualitative Overview:** Understand large data sets at a glance.
- **Pattern Discovery:** Search for trends, structures, irregularities, and relationships.
- **Analysis Support:** Find interesting regions and parameters for quantitative analysis.
- **Validation:** Provide visual proof of computer-derived representations.

2 Categorization of Techniques

1. **Pixel-Oriented:** Mapping dimension values to colored pixels.
2. **Geometric Projection:** Using geometric transformations and projections.
3. **Icon-Based:** Using features of icons (faces, sticks) to represent data.
4. **Hierarchical:** Partitioning subspaces into hierarchies.
5. **Complex Relations:** Visualizing networked or multi-relational data.

3 Technique Deep-Dive

3.1 Pixel-Oriented Visualization

For a dataset with m dimensions, m windows are created on the screen.

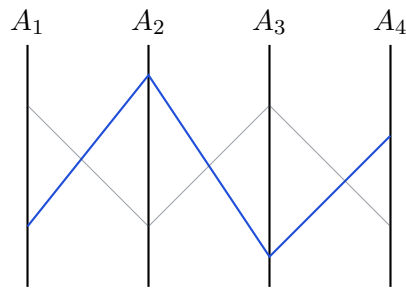
- Each record's m values are mapped to pixels at fixed positions.
- **Color Coding:** The pixel's color reflects the data value (e.g., higher income $\rightarrow$ brighter color).
- Often used for: Income, Credit Limit, Transaction Volume, Age.

3.2 Geometric Projection

Transforms multidimensional data to lower-dimensional visual representations.

Geometric Methods

- **Scatterplot Matrices:** 2D grids of scatterplots for every attribute pair.
- **Landscapes:** Visualization of geometric transformations as a 3D surface.
- **Parallel Coordinates:** Attributes are represented as parallel axes.
- **Projection Pursuit:** Finding "meaningful" 2D projections.
- **Hyperslice / Prosection Views:** Interactive multidimensional viewing.



Parallel Coordinates: Multidimensional point as a polyline.

3.3 Icon-Based Visualization

Maps data values to the visual features of icons.

- **Chernoff Faces:** Facial features represent variables.
 - **10 Characteristics:** Head eccentricity, eye size, eye spacing, eye eccentricity, pupil size, eyebrow slant, nose size, mouth shape, mouth size, and mouth opening.
- **Stick Figures:** Uses the angle/length of limbs to encode data.
- **General Techniques:**
 - **Shape Coding:** Using shapes to represent information encoding.
 - **Color Icons:** Using colored icons to encode additional information.
- **Tile Bars:** Small icons representing feature vectors in document retrieval.

3.4 Hierarchical Visualization

Uses hierarchical partitioning of subspaces to organize data.

- **Dimensional Stacking:** Embedding coordinate systems within others.
- **Tree-Maps:** Recursive rectangular partitioning based on hierarchy.
- **Worlds-within-Worlds:** n -dimensional visualization using nested 3D spaces.
- **InfoCube / Cone Trees:** 3D hierarchical layouts.

4 Proximity: Similarity and Dissimilarity

Proximity

A general term referring to either **Similarity** or **Dissimilarity** between objects.

Measure	Correlation with A likeness	Typical Range
Similarity	Higher value = More alike	$[0, 1]$
Dissimilarity	Lower value = More alike	$[0, \infty)$ (Min is 0)

4.1 Key Nuances

- **Similarity (s):** A numerical measure of likeness. $s = 1$ usually means identity.
- **Dissimilarity (d):** Often expressed as "distance". $d = 0$ means identical objects.

Visualizing proximity helps identify clusters and outliers in high-dimensional spaces.